Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Ali Shakouri	Email	coria-consortium@purdue.edu
Institute Name	Purdue University			
Address	1205 Mitch Daniels Boulevard, West Lafayette, Indiana 47907			

2. Technology Overview

Title of Proposed Technology	**Development of an Edge-Based AIoT Platform for Integrating Diverse Smart Devices and Creating Customized Automation Scenarios through Natural Language**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Information & Communication		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	U-Computing	U-Computing Platforms and Application Technologies	Home Network	Home Network applications and Service Technologies
Concept of Proposed Technology	**Core Concept** ● Market Expansion of Intelligent AIoT ■ Intelligent AIoT is an emerging technology that integrates AI capabilities into traditional IoT systems. ■ Global companies such as Amazon, Google, Apple, and Samsung are rapidly developing proprietary cloud-based platforms and expanding their device ecosystems. ■ According to Research and Markets ("AIoT Solutions by Applications and Services, 2023–2028"), the global AIoT market is expected to grow at a CAGR of 40.6%, reaching			

USD 91.2 billion by 2028.

- Limitations of Existing Cloud-Based AIoT Platforms
  - Current AIoT platforms employ a device-oriented control model, in which users must directly control individual devices.
    - IoT applications must be reconfigured each time devices are added or replaced.
  - Interoperability is limited; devices connected to different ecosystems (Apple, Google, Samsung, etc.) are difficult to integrate.
  - IoT data uploaded to the cloud raises significant concerns regarding privacy breaches.
  - Growing social concerns exist about smart-device hacking and security vulnerabilities.
- Addressing These Challenges with a Service-Based Edge AIoT Platform
  - The proposed platform adopts a service-based architecture, where users request services and devices are controlled automatically—similar to how an OS manages hardware in computing.
    - No need to modify IoT applications when devices are added or changed.
    - Enables seamless interconnection across devices from different manufacturers (e.g., Google, Apple, Samsung, Xiaomi).
  - By collecting data at the edge, privacy risks associated with cloud transmission are eliminated.
    - The system is designed to be resilient to device failures and hacking.
    - Edge nodes mediate device-to-device communication so devices cannot directly affect each other.
    - Local servers perform AI-based IoT big-data analysis to monitor cyber-intrusion in real time.
  - Introduces agentic AI for IoT: autonomous AI agents capable of sensing, decision-making, and executing tasks based on environmental data.

Key Functions and Potential Applications

Key Functions

- AI-Agent–Based User Interface (Figure 1(a))
  - Users do not manually control devices; instead, they specify the required functions (services) for each IoT application using a script-based interface.
  - Devices are registered and automation scenarios are defined through a mobile application.
  - LLM technologies translate natural-language descriptions of IoT automation rules into JOI-understandable script programs.
- Execution of the JOI Platform on Edge Devices (JOI Hub) (Figure 1(b))
  - The JOI Hub automatically discovers and selects appropriate devices to deliver specified services.
  - Supports a wide spectrum of communication protocols including Wi-Fi, BLE, Zigbee, Thread, and LoRa.

- Fully supports the Matter interoperability protocol.

- Interconnection of Diverse Devices Through the JOI Platform (Figure 1(c))
  - Devices operate in the sequence defined by the script program.
  - The JOI platform functions as the operating system of IoT, controlling devices based on user-defined automation programs.
  - Integrates smart devices with robots and cloud services, serving as a software infrastructure for Physical AI.
  - The platform can also be executed in the cloud for scalable deployment.
- Local Server–Based IoT Data Analytics (Figure 1(d))
  - Detects device failures and cyber-intrusion in real time.
  - Analyzes device-usage patterns to generate optimized automation algorithms (e.g., energy-saving operational recommendations).

(a) (b) (c) (d)

Figure 1. Architecture of the JOI AIoT Platform

Potential Applications

- Cost Reduction in Managing Diverse Sensors and Smart Devices
  - Applicable to smart buildings, smart farms, and smart factories.
  - Reduces the overhead of device replacement and upgrades by providing a unified platform.
  - Enables easy addition of new AIoT applications for energy management, security, air-quality control, and spatial optimization.
  - Supports integration of robots and smart devices to enable Physical AI.
  - Expands automation capabilities by abstracting cloud services as virtual devices.
- Secure AIoT Solutions for Privacy-Sensitive Environments
  - Applicable to military facilities, hospitals, and high-security institutions.
  - Enables secure smart-space deployment using local networks and the JOI Hub without relying on the internet or cloud.
  - Provides continuous monitoring of device failures and hacking attempts through local

	server analytics. ● Personalized Smart-Space Experiences Through Enhanced User Convenience ■ Suitable for smart homes, retail stores, and smart offices. ■ Maximizes user choice by integrating devices across different brands and communication protocols. ■ Allows users to easily create new automation scenarios using natural language. ● Additional AI-Agent Services and AI Service Development ■ Applicable to all institutions and users adopting the JOI platform. ■ Supports development of an open community for sharing user experiences and automations. ■ Provides opportunities for developing add-on AI services tailored to specific needs. (e.g., personalized pet-care assistants, plant-care management, spatial AI assistants, customized entertainment services)
Rationale for Joint R&D Project	**Critical Technical Barriers** ● Need for Advancing Situation-Awareness Technology for Agentic AI ■ Current AIoT automation still requires users to explicitly specify detailed automation scenarios. Although LLM-based natural-language automation is state-of-the-art, users remain responsible for precisely describing the service they need. ■ Future AIoT systems must evolve into agentic AI, where accumulated IoT data enables autonomous generation and recommendation of optimal automation scenarios even without explicit user requests. ■ The foundational capability of agentic AI is accurate recognition and interpretation of the state of the user and the surrounding environment. ● Necessity of Sensing Technologies for Monitoring Users, Pets, and Other Living Entities ■ Although various sensors exist for monitoring indoor environmental conditions, very few sensors can detect the conditions or behaviors of users or companion animals—who are the actual actors within the space. ■ Existing commercial sensors have limited functionality and are typically sold as bundled packages tied to proprietary smartphone apps. ■ Continuous monitoring is required not only indoors but also outdoors. ■ For agentic AI to function, it is essential to integrate multimodal information about users and animals with other smart devices in the environment. ■ Therefore, there is a strong need for the development of multimodal sensing technologies that can accurately monitor the states of humans and companion animals. **Why Global Collaboration?** ● Building a Differentiated Value Chain through Global Integration of Foundational and Applied Technologies ■ Demand is rapidly increasing for automated services tailored to individuals requiring care, as well as for companion animals.

(max 10 pages)

	■ Purdue University laboratories already possess expertise in monitoring cattle health using diverse sensor technologies, and this capability can be extended to detecting human and companion-animal health conditions and abnormal behaviors. ■ The joint R&D effort aims to develop next-generation smart-home solutions by integrating human- and pet-state sensing technologies with smart devices, enabling entirely new automation applications that are currently not achievable. ■ With the edge-based AIoT platform enabling seamless interoperability across diverse brands and protocols—and supporting natural-language automation—the project is poised to deliver an advanced home-solution ecosystem and expand into the U.S. market.
Key Words	1. Service-Oriented AIoT Platform 2. Edge AIoT Platform 3. Agentic AI 4. Multimodal Sensing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (7)
R&D Objectives	**R&D Objectives** The goal of this project is to integrate the global institute's multimodal sensing–based technologies for monitoring the health and behavior of living entities with an AIoT platform enhanced by agentic AI capabilities. Through IoT big-data analysis and learning, the platform will autonomously generate beneficial automation scenarios for users—even without explicit user requests—thereby enabling a next-generation AIoT system with proactive intelligence. **Detailed Objectives** 1. Development of Multimodal Sensing Technologies for Monitoring Human and Companion-Animal Health and Behavior (Purdue University) ● Development of privacy-preserving sensing methods for human monitoring. ● Integration of nonintrusive biosensors with anomaly-detection sensors that users or animals will not resist wearing. ● Incorporation of communication modules capable of interacting with the edge hub in offline (no-internet) environments. ● Ultra-low-power sensing technologies for wearable sensors. 2. Situation Recognition and Response Through Multimodal Sensor Data Analysis (MysMax) ● Aggregation of IoT data within a local server for sensor-fault and error analysis. ● Multimodal data analytics for anomaly detection. ● Automated generation of responsive automation scenarios based on detected anomalies. 3. Development of Agentic AI Capable of Delivering Beneficial Automation Scenarios Without Explicit User Requests (MysMax)			

- Learning-based modeling of user patterns using IoT data.
- Autonomous recommendation of beneficial automation scenarios even in the absence of user input.
- Expansion toward a Physical AI platform by integrating robots with smart spaces.
- Development of a software module that abstracts robots as service entities within the platform.

R&D Target Specifications

Key Performance Indicators	Target	Weight	Measurement and Verification Methods
Sensing accuracy	≥ 90%	30%	Create more than 100 test scenarios to e valuate anomaly-detection accuracy.
Battery autonomy	7 days	20%	Measure the operational lifetime of weara ble sensors without recharging.
Anomaly Detection L atency	< 5 sec	20%	Measure the total latency from sensing to analysis to user notification for 10 types of abnormal conditions.
Action Latency	< 5 sec	15%	Measure the maximum response time for automation scenarios triggered by more t han 10 types of abnormal events.
Number of sensor ty pes	≥ 3개	15%	Identify and document the types of senso rs used to monitor the state of humans a nd companion animals, including both we arable sensors and environmental sensor s.

R&D Plan

1. Milestones

Year	Key Objective	Major Tasks	Expected Outcomes
Year 1	Core Technology Development	Develop multimodal wearable sensors for monitoring the condition of humans and companion animals - Develop foundational technologies for agentic AI–based scenario generation	Secure low-power, low-cost, high-efficiency sensing technologies for humans and companion animals
Year 2	Platform Integr ation and Vali dation	Integrate wearable sensors with the AIoT platform - Automatically generate scenarios based on IoT data	Implement autonomous scenario-generation capabilities based on user and companion-animal status
Year 3	Mass-Production Readiness and Field Demonstration	- Validate and refine system performance through testbeds in the U.S. and Korea - Advance sensing and agentic AI technologies (power efficiency improvement and latency reduction)	Develop an MVP-level AIoT platform with automation functions for individuals and companion animals requiring care

2. Research approach and strategy

- Building an Integrated Value Chain:

  Establish a unified value chain by combining Purdue University's foundational wearable-sensing technologies with MysMax's AIoT platform capabilities, thereby expanding the applicability and impact of user-tailored automation scenarios.

- Testbed-Driven Validation:

  Set up testbeds in both the United States and Korea from the early stages of the project

to enable phased verification of research outcomes, ensuring continuous collaboration, communication, and joint experimentation between the two institutions.

- Market-Oriented Validation Strategy:

  Develop a validation framework from the outset that addresses U.S. smartphone and smart-home ecosystem entry barriers, ensuring that the technology aligns with the requirements and expectations of global target markets.

3. Research Approach and Methods

- [Sensor Development] Wearable Sensors

  Conduct research on multimodal sensing technologies capable of detecting the health status and abnormal behaviors of humans and companion animals, and develop corresponding wearable sensor systems.

- [Agentic AI Technology] Advancement of MysMax's AIoT Platform:

  Enhance the AIoT platform to analyze diverse sensor data on a local server and develop AI-driven capabilities that autonomously recommend automation scenarios without user intervention.

- [Integration] Unified Testbed Demonstration:

  Establish smart-home testbeds at Purdue University and in Korea to enable continuous validation, integration, and performance testing of the developed technologies under real-world conditions.

4. Research Plan

Research Activity	Year 1 (Q1-Q4)				Year 2 (Q1-Q4)				Year 3 (Q1~Q4)				Notes
Development of multimodal sensors for monitoring humans and companion animals													Led by Purdue
Development of automatic scenario-generation technology based on sensor data													Led by SME
Establishment and integration of testbeds in the U.S. and Korea													Joint Research
Platform evaluation													Link with end-users

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Role and Responsibilities of the Research Institute: Development and Implementation of Multimodal Sensor Technologies ● Technical Contribution and Research Scope ■ Development of multimodal sensing technologies capable of detecting health conditions and abnormal behaviors (e.g., falls) of humans and companion animals. ■ Development of edge-AI techniques that selectively transmit abnormal-event signals to in-home edge devices. ■ Implementation of ultra–low-power sensor technologies capable of operating for one week without recharging. ● Detailed Tasks ■ Development of multimodal wearable-sensor technology for detecting biosignals and abnormal behaviors of humans and companion animals. ■ Implementation of sensors equipped with communication modules that can transmit detected events from outdoor environments to indoor edge hubs. ■ Establishment of a smart home testbed in the United States to provide a real-world validation environment. ● Provided Resources ■ Sensor development equipment and research personnel for hardware and software implementation. ■ Infrastructure and facilities for building a smart home testbed.
Expected R&R of Korean SME Partner	Role and Responsibilities of the Korean SME Partner: Product Advancement and Validation for Applied Technology ● Technical Contribution and Development Tasks ■ Enhancement of the AIoT platform using agentic AI technologies, developing automated scenario-generation capabilities based on multimodal AIoT data. ■ Integration of the multimodal sensors developed by Purdue into the AIoT platform to enable accurate, context-aware automated responses for humans and companion animals. ■ Development of the integrated AIoT platform architecture. ● Integrated Testing and Verification Activities ■ Performance validation through testbeds: quantitative evaluation of Purdue's sensor performance (accuracy, operational duration, and communication latency). ■ Verification of end-to-end system performance, including anomaly detection time and scenario-response execution time. ■ Development of an MVP-level AIoT platform tailored for smart-home deployment in both the United States and Korea. ● Collaboration Framework and Resource Commitment ■ Provision of the AIoT platform to Purdue University to support testbed construction. ■ Establishment of a continuous collaboration framework through joint validation using testbeds.

(max 10 pages)

	■ Resource contributions: provision of the edge-based AIoT platform (user interface, edge devices, platform software, IoT data analytics software) and deployment of expert engineering personnel.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. Technical Achievements ● Development of low-power multimodal sensing technology for detecting the condition of humans and companion animals ■ Established a foundational technology that expands the range of detectable anomaly states through multimodal sensor analysis by integrating Purdue University's wearable sensing technology with MySMAX's smart-device integration technology. ■ Built a multimodal sensor data database that can support future studies and applications. ● Advancement of the service-oriented AIoT platform using agentic AI ■ Enhanced the existing AIoT platform—which controlled devices through user-defined natural-language automation scenarios—into a system that can autonomously recommend beneficial automation scenarios without explicit user requests. ■ Achieved the world's first implementation of specialized scenario-generation technology that adapts to the detected state of humans and companion animals as well as the status of installed smart devices. 2. Market Potential ● Securing global competitiveness through smart-home technologies tailored for households with care-dependent individuals or companion animals ■ Differentiation through personalized features that allow users to easily add automation scenarios. ■ Expanded technological gap with competing AIoT platforms by selectively activating scenarios based on the condition of humans or companion animals. ● Market expansion through scalable technology ■ Extends beyond smart homes to broader smart-space applications such as hospitals, factories, and buildings based on validated core features. ● Maximizing profitability and reducing operational costs ■ Integration of multimodal sensors reduces the cost associated with purchasing and consolidating separate sensors. ■ Embedding sensors directly into the AIoT platform reduces latency and protects user privacy. 3. Industrial and Economic Impact ● Market expansion through technological and cost innovation ■ A single AIoT platform integrates diverse brands and protocols of smart devices and supports a wide range of automation scenarios, reducing the need to purchase multiple separate solutions. ■ Lower smart-space deployment costs broaden the user base. ● Energy savings and enhanced safety (ESG leadership)

(max 10 pages)

	■ Builds optimal smart environments that improve both energy efficiency and safety based on the condition of humans and companion animals. ■ Supports carbon-neutral goals by optimizing space-level energy use through smart-space automation. ● Strengthening national technological leadership in strategic domains ■ Through substantive collaboration with Purdue University, the project aims to elevate Korea's national strategic competencies in smart-home and AI technologies to a globally leading level.

6. References

(1) Euiseok Kim, Taehyeong Son, Soonhoi Ha, "A novel hierarchical edge-based architecture for service oriented IoT," Internet Of Things, Sep, 2023.

(2) https://www.startupcityapac.com/mysmax, "MySmaX, Transforming Smart Spaces through Innovative AIoT Solutions", 2025

(3) Yigit Tuncel, Toygun Basaklar, Mackenzie Smithyman, João Dórea, Vinícius Nunes De Gouvêa, Younghyun Kim and Umit Ogras, "Advancing Cattle Welfare: Ultra Low-power Health Monitoring at the Edge" Biomedical Circuits and Systems Conference (BioCAS), 2023